Experiment No: 4			
Course Name:	Web Programming Laboratory (216U01L405)	**Semester:**	**IV**
Date of Performance:	**06/ 03/ 2026**	**DIV/ Batch No:**	**B2**
Student Name:	**Ashwera Hasan**	**Roll No:**	**16010124107**

Title: Develop a JavaScript program that implements client-side validation for HTML forms.

Aim of the Experiment:

- The aim of this project is to **develop a JavaScript program that implements client-side validation for HTML forms**, enhancing user experience by providing immediate feedback and reducing server-side processing.

Objectives for the Experiment:

1. **Implement Basic Validation**:
   - Check for required fields and ensure they are filled before submission.
   - Validate data types (e.g., ensuring email fields contain valid email addresses).
2. **Enhance User Feedback**:
   - Provide clear error messages that guide users on how to correct their input.
   - Highlight fields that contain errors dynamically.
3. **Integrate Advanced Validations**:
   - Implement complex validations (e.g., password confirmation, minimum/maximum length).
   - Consider real-time validation as the user types.
4. **Test and Debug the Program**:
   - Use various test cases to ensure that all validation scenarios are covered.
   - Fix any issues to ensure smooth user interaction.
5. **Document the Code**:
   - Write thorough documentation explaining the code, functions, and intended behavior of the validation process.

COs to be achieved:

Design dynamic web pages using various HTML tags.
Use CSS to prepare the layout of web pages.
Apply JavaScript for validation in client-side programming.

Books/ Journals/ Websites references:

1. Students should write

Theory:
Students should write about CSS3 (citation needed for content taken (IEEE or APA7 style))

Problem statement: "College Event & Student Services Web Portal" A college wants to develop a basic HTML5-based web portal to showcase events, registrations, information, and services for students. Each task contributes one page or feature of this portal. Enhanced the portal using CSS3 and Java script with validation.

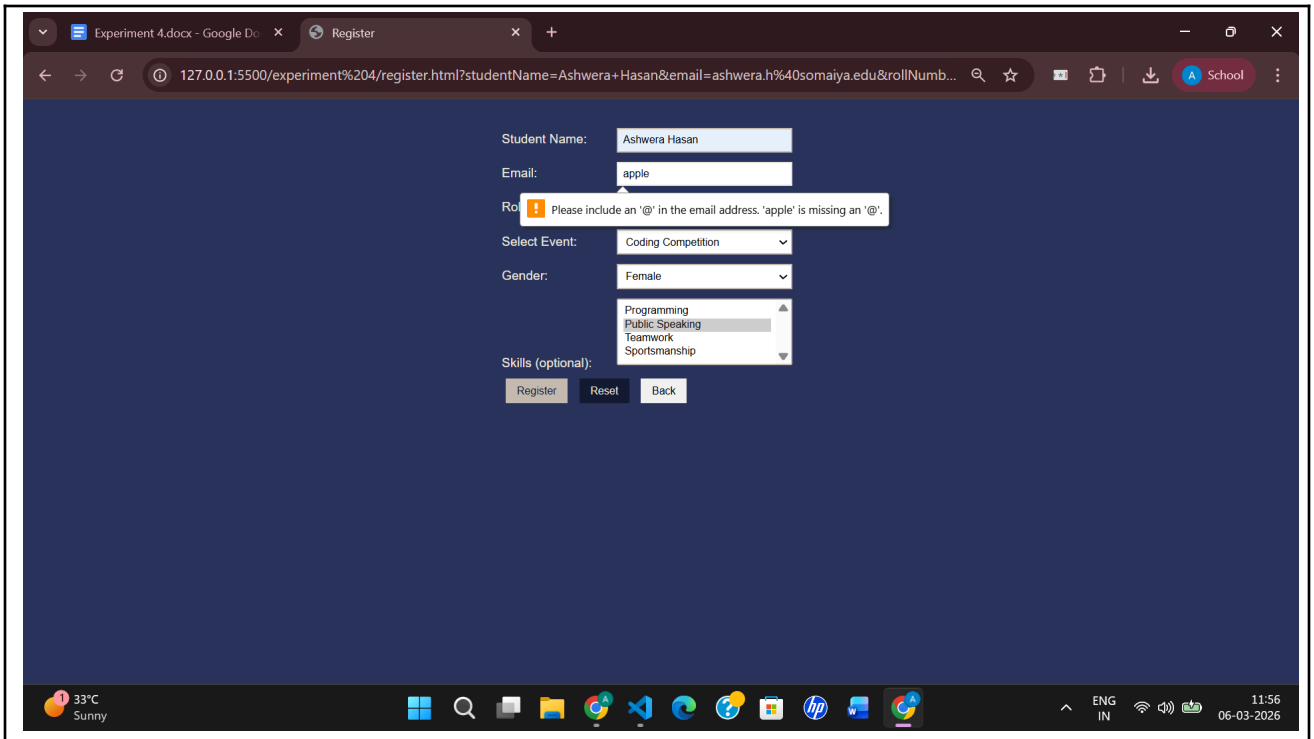
Repository: https://github.com/ashwera/webProgrammingLab

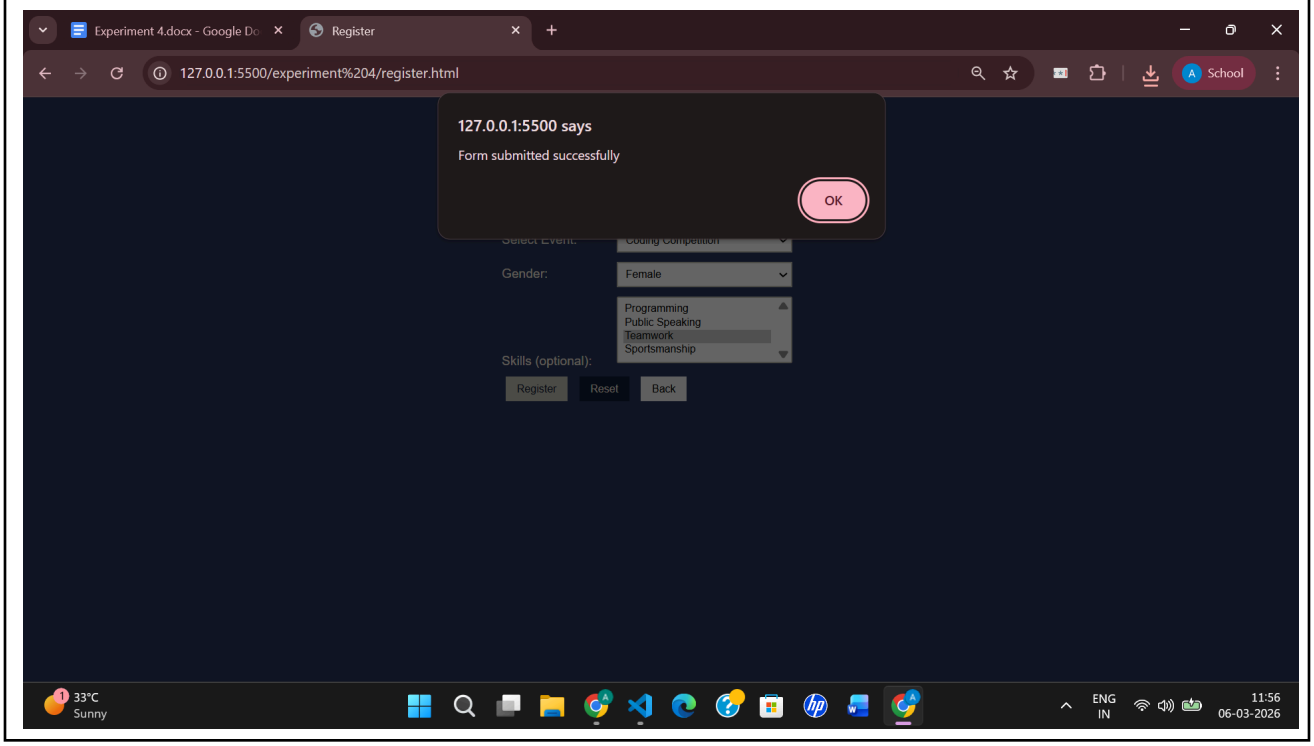
Code:

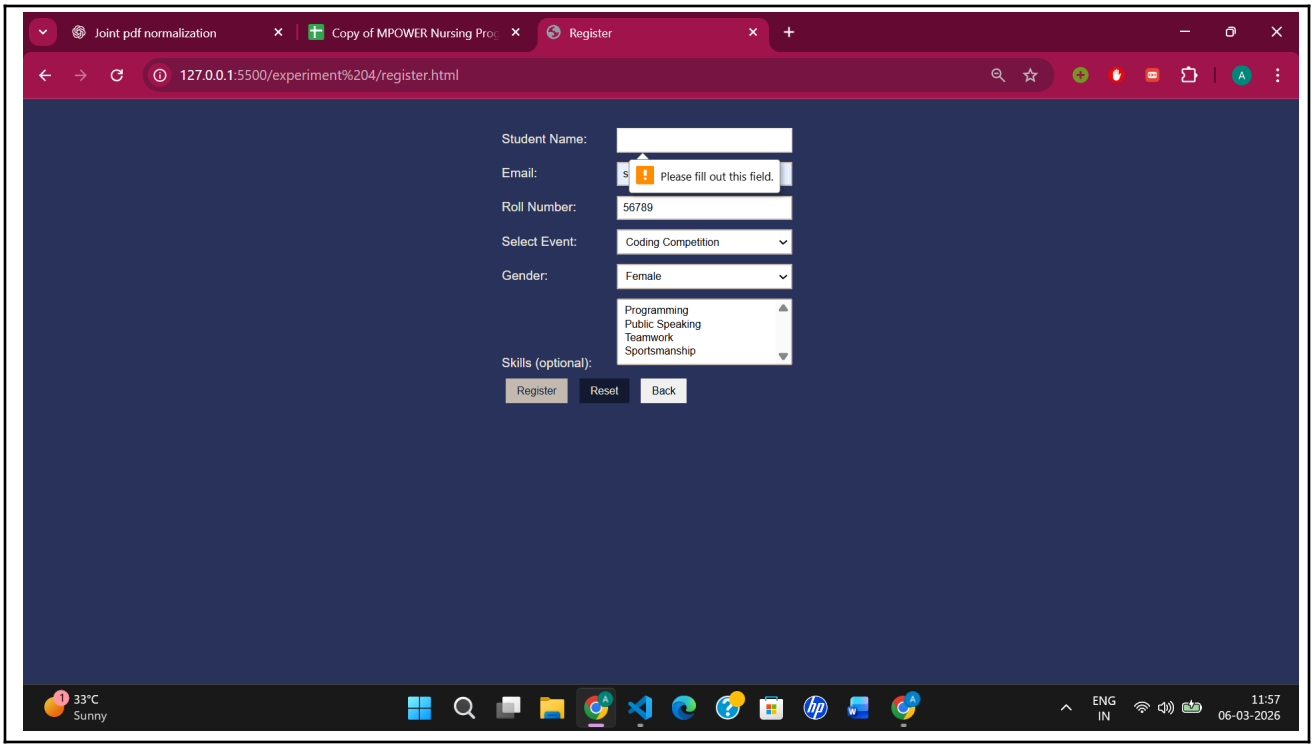
```javascript
document
  .getElementById("registrationForm")
  .addEventListener("submit", function (e) {
    let name = document.getElementById("studentName").value.trim();
    let email = document.getElementById("email").value.trim();
    let roll = document.getElementById("rollNumber").value.trim();
    let event = document.getElementById("eventSelection").value;
    let gender = document.getElementById("genderSelection").value;

    let namePattern = /^[A-Za-z ]+$/;
    let emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

    if (name === "" || !namePattern.test(name)) {
      alert("Enter valid name");
      e.preventDefault();
      return;
    }

    if (!emailPattern.test(email)) {
      alert("Enter valid email");
      e.preventDefault();
      return;
    }
```

```
    if (roll.length < 3) {
      alert("Roll number too short");
      e.preventDefault();
      return;
    }

    if (event === "") {
      alert("Select an event");
      e.preventDefault();
      return;
    }

    if (gender === "") {
      alert("Select gender");
      e.preventDefault();
      return;
    }

    alert("Form submitted successfully");
  });
```

Output / Screenshot:


Experiment 4.docx - Google Do
Register
127.0.0.1:5500/experiment%204/register.html?studentName=Ashwera+Hasan&email=ashwera.h%40somaiya.edu&rollNumb...
School
Student Name:
Ashwera Hasan
Email:
apple
Please include an '@' in the email address. 'apple' is missing an '@'.
Select Event:
Coding Competition
Gender:
Female
Programming
Public Speaking
Teamwork
Sportsmanship
Skills (optional):
Register
Reset
Back
33°C
Sunny
ENG
IN
11:56
06-03-2026


Experiment 4.docx - Google Do
Register
127.0.0.1:5500/experiment%204/register.html
School
127.0.0.1:5500 says
Form submitted successfully
OK
Gender:
Female
Programming
Public Speaking
Teamwork
Sportsmanship
Skills (optional):
Register
Reset
Back
33°C
Sunny
ENG
IN
11:56
06-03-2026



Post Lab Subjective/Objective type Questions:

1. What are potential downsides of relying solely on client-side validation?
   Client-side validation can be bypassed by disabling JavaScript or modifying the code. Therefore, server-side validation is necessary for security and data reliability.
2. How would you modify your validation logic to accommodate accessibility for users with disabilities?
   We modify our validation logic by using accessible error messages and labels for form fields. Also provide clear text feedback and ARIA attributes so screen readers can understand validation errors.
3. What strategies could you use to manage different validation scenarios for various form types (e.g., login, registration)?
   Different validation scenarios can be managed by creating separate validation functions for different form types such as login or registration. Each function can apply specific rules based on the required fields and input formats needed for that particular form.
4. How can you ensure that your validation mechanisms are up to date?
   By regularly reviewing and updating validation rules based on new requirements and security practices, we can ensure that our validation mechanisms are updated. Testing and maintaining the code helps ensure the validation mechanisms remain accurate and effective.

Conclusion and Discussion:

Client-side form validation using JavaScript was successfully implemented. It checks user input before submission and provides immediate feedback. This improves user experience and reduces invalid data being sent to the server.

Feedback:

1. Riya: Good and well structured script
2. Mohammad: Script looks properly formatted
3. Krish: The javascript code works as intended.